

Wellness Lodge

Website and reservation platform architecture

Proposed structure, booking rules and operating controls for client review

STATUS

For client review

PREPARED FOR

Hotel or lodge management

DATE

5 September 2026

1 Purpose and recommendation

The proposed solution combines the public website, booking engine and an installable front desk application in one system. Firebase will host the applications and provide the shared database, staff authentication, file storage, notifications and server functions. GitHub will hold the source code and deployment history. The public website will use the client's .com.pg domain.

Approval of the booking rules should come before visual design. The rules determine when a room is available, how long it may be held, who can confirm a transfer, and what happens when staff do not act on a request.

1.1 RESPONSIBILITY BOUNDARIES

The Project Coordinator will prepare and lodge the payment gateway and domain applications on the client's behalf. Lodge management remains responsible for supplying accurate business records, approving commercial rules, signing declarations, paying third party charges and responding to requests from the bank or registry. Submission does not guarantee approval. Kina Bank, BSP and PNGUoT make their own assessment and may change their requirements, fees or processing time.

Party	Responsibility
Lodge management	Provide complete business information and documents; approve rates, policies and booking rules; nominate authorised signatories; pay bank, domain and other third party charges; approve the final system.
Project Coordinator	Prepare application packs, explain missing items, lodge the payment gateway and .com.pg applications, follow up with the institutions and keep management informed.
Developer	Build and test the website and front desk app; configure Firebase, DNS and approved payment integration; provide technical details requested by the bank or registry.
Front desk and accounts	Confirm operating procedures, test daily workflows, verify transfers, reconcile payments and complete guest follow-up tasks after training.
Bank and PNGUoT	Assess applications, request further documents, set commercial or technical conditions, issue credentials or domain approval, and confirm production readiness.

2 Project scope

Area	Included in the proposed system
Public website	Property pages, accommodation, car rental, restaurant, cafe, function hall, catering, events, past events gallery, facilities, policies and contact details.
Reservations	Live availability, date selection, room choice, guest details, pricing, deposit rules, confirmation and booking reference.
Payments	Online gateway payment, direct bank transfer with receipt upload, and pay at front desk requests.
Front desk	Installable application, live booking queue, alerts, calendar, payment review, guest follow-up and audit history.
Administration	Rooms, rates, seasonal pricing, blocked dates, users, roles, policy content, reports and system settings.

3 Users and access

User	Main access
Guest	Search dates, reserve, pay, upload a receipt, receive notifications and view a booking through a secure link.
Front desk officer	Review requests, accept or reject bookings, verify bank transfer evidence, contact guests, check guests in and complete tasks.
Manager	All front desk functions plus rates, overrides, refunds, reports and escalation review.
Content or pricing administrator	Maintain approved rates, rooms, vehicles, menus, services, policies and website content within assigned permissions.
Administrator	User accounts, roles, property configuration, integrations, audit access and system settings.

4 Booking lifecycle

Each reservation will have one clear status. Availability changes only through secure server logic, not through the calendar displayed in the guest's browser.

Status	Meaning	Inventory effect
Draft	Guest has started but not submitted the booking.	No lock
Pending payment	Gateway payment has started but is not verified.	Short checkout hold, subject to agreed duration
Awaiting receipt	Guest selected bank transfer and submitted the request or receipt.	No lock unless management approves a temporary hold
Awaiting front desk	Staff action is required.	No lock until accepted
Held	Front desk accepted a pay at desk request.	Locked for 12 hours
Confirmed	Payment is verified or an authorised staff member confirmed the booking.	Locked for the full stay
Expired or cancelled	The hold ended or the booking was cancelled.	Dates released
Checked in or completed	The guest arrived or the stay is complete.	Historical record retained

5 Payment and reservation rules

5.1 ONLINE GATEWAY

The room is confirmed only after the bank or gateway sends a valid server to server payment result. A browser success page alone is not enough. The server records the transaction reference, amount, currency and verification result, then locks the dates and sends confirmation to the guest and front desk.

5.2 DIRECT BANK TRANSFER

The guest receives the lodge bank details, booking reference, amount and payment deadline. The booking page will provide a secure receipt upload and this will be the official submission channel. Email to payments@wellnesslodge.com.pg will remain available as a fallback, but the reservation must not depend on the email inbox having free space. Staff must compare the uploaded receipt with the bank account before marking the payment as verified. Once confirmed, the dates are locked.

5.3 RECEIPT UPLOAD CONTROLS

- Accept only approved image and PDF formats and apply a reasonable file-size limit.
- Scan or validate uploads, reject unsafe files, and store accepted receipts outside the public website area.
- Attach each receipt to one booking reference and record the guest, amount stated, upload time and current review status.
- Show the receipt immediately in the front desk payment queue and issue the normal visual, sound and push alerts.
- Allow authorised staff to verify, reject or request a clearer receipt. Record the staff member, action, reason and time.
- Treat the receipt as evidence only. Payment becomes verified after staff confirms the funds in the lodge bank account.
- Send the guest an acknowledgement after upload, then a separate confirmation or rejection after review.
- Apply an approved retention period and remove expired receipts according to management policy and legal requirements.

5.4 PAY AT FRONT DESK

A pay at front desk request enters the staff queue. Once accepted, the system creates a 12 hour hold and displays the exact expiry time. If staff do not convert the hold to a confirmed booking before expiry, a scheduled server process releases the room automatically.

6 Availability and double booking control

- The server checks every date in the requested stay immediately before creating a hold or confirmation.
- A database transaction writes the reservation and inventory lock together. If another booking wins the same dates first, the second request is declined and the guest is asked to choose again.
- Changes, cancellations and expiry events update the room inventory through the same controlled service.
- Staff overrides require a reason and remain visible in the audit history.

7 Technical structure

Component	Proposed technology	Responsibility
Public website	Responsive web application on Firebase Hosting	Marketing pages, availability search, checkout and guest booking access.
Front desk app	Installable Progressive Web App	Live operations, alerts, payment review, calendar, task list and guest management.
Management portal	Secure web administration application	Rates, inventory, services, website content, staff permissions, publishing and change history.

Component	Proposed technology	Responsibility
Database	Cloud Firestore	Rooms, rates, reservations, payments, tasks, users, notifications and audit records.
Secure services	Firebase Cloud Functions	Availability checks, payment verification, expiry, email, alerts and privileged actions.
Identity	Firebase Authentication	Staff sign in, password reset, multi factor option and role control.
Files	Cloud Storage	Transfer receipts and property media, protected by access rules.
Notifications	Firebase Cloud Messaging and email provider	New booking alerts, reminders, expiry warnings and guest messages.
Delivery	GitHub and controlled deployment workflow	Source history, review, testing and deployment to Firebase.

8 Front desk application

- New reservation queue with received time, guest, dates, room, payment method and response deadline.
- Daily arrivals, departures, current guests and no show list.
- Availability calendar with room blocks, maintenance periods and authorised overrides.
- Bank transfer receipt viewer with verify, reject and request more information actions.
- Outstanding deposits and balances, with payment history and receipt references.
- Follow-up tasks for calls, email, transport, arrival time, room preparation and balance collection.
- Management reporting for occupancy, booking source, revenue, cancellations and response time.

8.1 ADDITIONAL LODGE SERVICES

The public website will include the car rental, restaurant, cafe and function hall. Management must also confirm whether the lodge provides outside catering, event planning or event support. Each service may be presented for information and enquiries, offered as an extra during room booking, or operated through a separate reservation process. Vehicle availability, table reservations, food ordering, catering orders and function hall calendars require their own rules and may change the development scope and cost.

A Past Events page will showcase approved weddings, meetings, conferences, celebrations and catered events. Management must approve each photograph and description and confirm that the lodge has permission to publish identifiable people, client names, testimonials and event details.

During discovery, management will provide a complete inventory of all current and planned services and activities, including those delivered through outside operators. Each item will be classified as information only, enquiry, quotation, accommodation add-on or separate online reservation. Only services selected for the approved first release will be built into transactional workflows.

8.2 MANAGEMENT ADMINISTRATION PORTAL

The system will include a secure management portal so authorised staff can maintain prices and business information without editing source code. The portal is separate from the public website and from routine front desk actions. Access will be controlled by role, and sensitive changes will remain traceable.

Administration area	Management capability	Control
Room inventory	Create and update room numbers, categories, capacity, beds, amenities, descriptions, images and sale status.	Removing a room cannot erase historical bookings.
Room rates	Maintain standard, weekend, seasonal, corporate, group and promotional rates, including taxes, deposits and extra guest charges.	Rates use start and end dates; overlapping or incomplete rate periods are rejected.
Other services	Maintain car rental, menus, function hall, catering, events, activities, packages, opening hours, prices and availability.	Transactional features remain limited to the approved scope.
Website content	Edit approved business profile, contacts, policies, galleries, Past Events, frequently asked questions and service information.	Draft and preview before publishing; publication rights may be restricted.
Users and alerts	Invite staff, assign roles, deactivate access and maintain notification or escalation settings.	Only administrators can grant privileged access.
Audit and recovery	Review changes and restore an earlier approved value where supported.	Record the user, time, old value, new value and reason for sensitive changes.

Every confirmed reservation stores a price snapshot containing the rate, taxes, deposit and extras accepted at booking time. A later rate change applies to new searches and bookings only. It must not silently recalculate an existing confirmed reservation. An authorised manager may amend an existing booking through a separate action that records the reason and customer communication.

9 Alerts and unattended requests

A new request will appear immediately in the front desk queue. The app can play a ringtone while it is open, show a visual alert, update its unread count and send a push notification to approved devices. Staff must first acknowledge the alert, then complete an action such as accept, reject, verify payment or request more information.

Modern browsers restrict sound when an application is closed or has not received notification and audio permission. The installation process will therefore include a notification test and a sound test on every front desk device. Persistent background alarms on every phone cannot be guaranteed by a browser application. If that behaviour is mandatory, a native Android or Windows wrapper should be assessed.

Elapsed time	Proposed response
Immediately	In-app ringtone, visual notice and push notification.
Not acknowledged	Repeat in-app alert at the agreed interval.
Acknowledged but not actioned	Keep the item in the outstanding queue and show its age.
Escalation threshold reached	Notify the manager or second nominated staff member.

10 Security and records

- Guests will never enter card details into the lodge database. The bank hosted or bank approved payment page handles card information.
- Staff access will follow assigned roles. Sensitive actions require authentication and are written to the audit log.
- Transfer receipts will use private storage rules, time limited access links, upload validation and role based access.
- The system will validate amounts, booking references and payment callbacks on the server.
- Backups, retention periods, privacy wording and incident contacts will be agreed before launch.
- Firebase App Check, security rules, logging and rate controls will reduce automated abuse.

11 Delivery stages

Stage	Output	Client decision
1 Discovery	Approved rooms, rates, policies, roles and booking rules.	Sign off the operating rules.
2 Design	Site map, screen designs, content plan and tested booking flow.	Approve the customer and staff experience.
3 Core build	Website, availability, reservations, front desk app and administration.	Review a working test system.
4 Payments	Gateway, bank transfer review, receipts and reconciliation.	Complete bank onboarding and test transactions.
5 Readiness	Security review, scenario testing, training and operating guide.	Approve launch readiness.
6 Launch	Firebase deployment, .com.pg connection and monitored release.	Authorise production launch.

12 Decisions required before build

- Whether inventory is managed by individual room, room category, or both.
- When the 12 hour hold begins and whether any shorter hold applies during gateway checkout.
- Deposit percentage or amount, payment deadline, cancellation terms and refund rules.
- Preferred gateway after Kina Bank and BSP confirm commercial and technical terms.
- Alert repeat interval, manager escalation time and staff operating devices.
- Who may override availability, price, expiry or payment status.
- Whether car rental is promotional, a room booking add-on, or a separate vehicle reservation system with availability and deposits.
- Whether the restaurant and cafe require menus only, table bookings, guest meal packages, takeaway ordering or online payment.
- Whether the function hall uses an enquiry form or a live event calendar with deposits, packages and booking confirmation.
- Whether the lodge offers outside catering, event planning, decorating, equipment hire or event support, and whether these services require quotations or online requests.

- Which past events may be published and who will approve photographs, captions, testimonials and client references.

13 Acceptance tests

- Two guests cannot confirm the same room for overlapping dates.
- A successful verified gateway payment locks inventory once and does not create a duplicate booking when the callback repeats.
- A failed or abandoned payment does not remain confirmed.
- A 12 hour hold expires at the correct local time and releases inventory.
- A verified bank transfer locks the dates and records the staff member who confirmed it.
- A valid receipt upload remains available even when the payments email mailbox is full and appears in the authorised front desk queue.
- New and overdue requests appear on every authorised front desk device after reconnection.
- Staff without permission cannot change rates, issue refunds or override room locks.
- An authorised pricing administrator can schedule a future rate and the public website uses it only during the correct effective period.
- Changing a room or service rate does not alter the agreed price on an existing confirmed booking.
- The system rejects overlapping rate periods and records the previous and new values for approved changes.
- Draft website content is not visible publicly until an authorised publisher approves it.
- A deactivated staff account can no longer enter the administration portal or front desk application.